

TWG2 Manufacturing and Construction Technologies

INTRODUCTION

The United Republic of Tanzania, through the Ministry of Education, Science and Technology (MoEST), is developing the National Technology Roadmap (NTRM) 2050 to support the recently approved Tanzania Development Vision 2050 (Dira 2050). This questionnaire seeks your valuable input in identifying critical technologies that will transform Tanzania's Manufacturing and Construction sector and drive the nation towards becoming an industrialized, knowledge-based, upper-middle-income country with a USD 1 trillion economy by 2050. Your expertise in Manufacturing and Construction is crucial for identifying technologies that will: Transform Tanzania into a globally competitive manufacturing hub, Increase manufacturing contribution from current 8.1% to a substantial share of GDP to 15%, Drive industrialization and economic diversification away from agriculture dependence, Create quality jobs and reduce dependence on imports through local value addition, Support urbanization with sustainable construction and smart city technologies for 50%+ urban population by 2050, Build resilient infrastructure to support 118+ million people.

SECTION A: RESPONDENT INFORMATION

A1. Please provide the following personal or organizational details for identification and communication purposes related to this assessment.

Name (optional)

HAPPINESS KAPANDA

Organization or institution

CONFEDERATION OF TANZANIA INDUSTRIES

Position or title

MEMBERSHIP OFFICER

Email address

membership@cti.co.tz

Phone number

0652313909

A2. Manufacturing and construction sub sector expertise

A2.1 Manufacturing expertise (select all that apply)

- ☐ Heavy manufacturing (iron, steel, machinery)
- ☐ Chemical and petrochemical manufacturing
- ☐ Pharmaceutical manufacturing
- ☐ Food and beverage processing
- ☐ Textile and garment manufacturing
- ☐ Leather and footwear manufacturing
- ☐ Wood and furniture manufacturing
- ☐ Paper and printing
- ☐ Plastics and rubber products
- ☐ Electronics and electrical equipment
- ☐ Automotive and transport equipment
- ☐ Metal fabrication and engineering
- ☐ Cement and building materials
- ☐ Glass and ceramics
- ☐ Industrial machinery manufacturing
- ☐ Consumer goods manufacturing
- ☐ Packaging industry
- ☐ Agro processing (industrial scale)
- ☒ Other

Other manufacturing expertise (specify)

Manufacturers Business Association

A2.2 Construction and infrastructure expertise (select all that apply)

- ☐ Residential construction
- ☐ Commercial construction
- ☐ Industrial construction
- ☐ Roads and highways construction
- ☐ Railway construction
- ☐ Port and airport construction
- ☐ Water and sanitation infrastructure
- ☐ Energy infrastructure (power plants, transmission)
- ☐ Telecommunications infrastructure
- ☐ Green building and sustainable construction
- ☐ Smart cities and urban planning
- ☐ Real estate development
- ☐ Building materials production
- ☐ Construction equipment and machinery
- ☐ Other

A2.3 Cross cutting expertise (select all that apply)

- ☐ Industrial engineering and automation
- ☐ Quality control and standards
- ☐ Industrial design
- ☐ Manufacturing management
- ☐ Construction project management
- ☐ Industrial technology transfer
- ☒ Industrial policy and planning
- ☐ Other

A3. Select scale of operations (if applicable)

- ☐ Micro enterprise (less than 5 employees)
- ☐ Small enterprise (5 to 49 employees)
- ☐ Medium enterprise (50 to 99 employees)
- ☐ Large enterprise (100 or more employees)
- ☐ Multinational operation
- ☒ Not applicable
- ☐ Other

A4. Geographic focus (select all that apply)

- ☐ Dar es Salaam
- ☐ Arusha
- ☐ Mwanza
- ☐ Mbeya
- ☐ Dodoma
- ☐ Zanzibar
- ☒ National coverage
- ☐ Regional or East African market
- ☐ Other

SECTION B: TECHNOLOGICAL GAPS AND TRANSFORMATION CHALLENGES

B1. What do you consider the three most critical technological gaps currently hindering development in Tanzania's manufacturing and construction sector?

Consider gaps in your specific sub sector

Gap

Outdated machineries and processes.

Why is this critical

Limits automation and smart manufacturing hence.

Current impact on productivity and competitiveness

Unfair competition and reduce productivity and quality

Gap

Limited technical infrastructure and connectivity to support technology.

Why is this critical

Makes it difficult to support new technology.

Current impact on productivity and competitiveness

Hinders efficient work flow and logistics.

Gap

Shortage of skilled technical personnel.

Why is this critical

Skills mismatch.

Current impact on productivity and competitiveness

Limits technology and innovation.

B2. Which of the following represents the greatest technological challenge for Tanzania's manufacturing and construction sector to achieve Vision 2050 goals? (Select at most three)

Select up to three options.

- ☐ Outdated manufacturing equipment and machinery
- ☐ Limited automation and Industry 4.0 adoption
- ☐ Inadequate quality control and testing facilities
- ☐ Poor industrial design and product development capabilities
- ☐ Limited local raw material processing capacity
- ☒ Insufficient skilled technical workforce
- ☐ Lack of modern construction technologies and techniques
- ☐ Inadequate industrial research and development infrastructure
- ☒ Limited access to affordable industrial financing
- ☒ Unreliable energy supply for manufacturing
- ☐ Poor industrial infrastructure (special economic zones, industrial parks)
- ☐ Weak linkages between manufacturing and raw material sectors
- ☐ Limited adoption of green and sustainable manufacturing practices
- ☐ Insufficient standardization and certification systems
- ☐ Other

SECTION C: MANUFACTURING AND CONSTRUCTION TECHNOLOGY IDENTIFICATION

Instructions: This section focuses on identifying and prioritizing technologies for manufacturing and construction. List up to three (3) priority technologies, then answer the detailed assessment questions for each technology.

How many technologies will you list? (1–3)

Type a number between 1 and 3, then tap Next.

1

*** C2. First Technology name**

Industrial Automation

*** C3. Technical feasibility in Tanzania context (1=Low, 5=High)**

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☒ 5

*** C4. Current readiness level in Tanzania (1=Not present, 5=Widespread use)**

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ 4
- ☐ 5

C5. Detailed technology assessment of Industrial Automation

*** C5.1 Technology description of Industrial Automation**

Describe the technology and the specific agricultural problem it solves.

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*** C5.2 Current status in Tanzania for Industrial Automation**

- ☐ Not present in Tanzania
- ☐ Research or pilot stage (specify institution)
- ☐ Limited commercial use (estimated number of facilities)
- ☒ Moderate adoption (market penetration percent)
- ☐ Widespread use

*** C5.3 Target beneficiaries (select all that apply) for Industrial Automation**

- ☒ Large manufacturing companies
- ☒ SME manufacturers
- ☒ Artisans and informal sector manufacturers
- ☐ Construction companies (large)
- ☐ Construction SMEs
- ☐ Real estate developers
- ☐ Infrastructure contractors
- ☐ Local raw material suppliers
- ☐ Industrial workers (job creation)
- ☐ Consumers (affordable products)
- ☐ Export markets
- ☐ Other

*** C5.4 Technology acquisition strategy for Industrial Automation (Select primary approach).**

- ☐ Develop locally through research and development (TIRDO, universities)
- ☐ Import turnkey equipment or plant
- ☐ Technology licensing from global manufacturers
- ☐ Joint venture with foreign technology partner
- ☐ Acquire through foreign direct investment (foreign company sets up)
- ☐ South to South technology transfer
- ☐ Reverse engineering and adaptation
- ☐ Public private partnership
- ☐ Special economic zone attraction
- ☒ Combination or other (specify)

C5.5 Infrastructure prerequisites (select all that apply) for Industrial Automation

- ☒ Reliable electricity (24/7)
- ☐ Three phase power supply
- ☐ Natural gas supply
- ☐ Water supply
- ☐ Waste treatment facilities
- ☐ Industrial land
- ☐ Proximity to port
- ☐ Rail connection
- ☐ Highway access
- ☐ Telecommunications or high speed internet
- ☐ Skilled labor pool nearby
- ☐ Industrial park or special economic zone
- ☐ Waste management systems
- ☐ Environmental compliance infrastructure
- ☐ Other

C5.6 Estimated total investment needed for Industrial Automation

- ☐ Less than USD 1 million
- ☐ USD 1 to 10 million
- ☐ USD 10 to 50 million
- ☒ USD 50 to 100 million
- ☐ USD 100 to 500 million
- ☐ More than USD 500 million
- ☐ Do not know

C5.7 Financing sources (select all that apply) for Industrial Automation

- ☐ Government industrial & infrastructure budget
- ☐ Development / Industrial bank financing
- ☐ Commercial banks (manufacturing & construction lending)
- ☐ Microfinance institutions
- ☐ Industry associations / SACCOs
- ☐ Private manufacturing & construction investment
- ☐ Development partners
- ☐ PPPs with manufacturing & construction firms
- ☐ Supply chain / project financing
- ☐ Green / climate infrastructure finance
- ☒ Other

If other, specify.

C5.8 Barriers to adoption (select all that apply) for Industrial Automation

Agricultural-specific barriers.

- ☐ High capital investment requirement
- ☐ Difficulty accessing industrial financing
- ☒ Limited local technical expertise
- ☐ Unreliable power supply
- ☐ High cost of electricity
- ☐ Import competition (cheaper imports)
- ☐ Small domestic market size
- ☐ Limited access to industrial land
- ☐ Complex licensing and permits
- ☐ High import duties on machinery
- ☐ Limited local supply chain
- ☐ Lack of industrial infrastructure
- ☐ Technology obsolescence risk
- ☐ Currency exchange rate volatility
- ☐ Limited export market access
- ☐ Poor road or port infrastructure
- ☐ Other

C5.9 Environmental and sustainability considerations (select all that apply) for Industrial Automation

- ☐ Low emissions or clean technology
- ☒ Energy efficiency systems
- ☐ Circular economy or recycling integration
- ☐ Wastewater treatment system
- ☐ Air pollution control equipment
- ☐ Solid waste management
- ☐ Noise reduction measures
- ☐ Carbon footprint reduction measures
- ☐ Other

C5.10 Value chain position (select all that apply) for Industrial Automation

- ☐ Raw material extraction or primary processing
- ☐ Intermediate goods manufacturing
- ☐ Final goods manufacturing or assembly
- ☐ Packaging
- ☐ Quality control or testing
- ☐ Construction or installation
- ☐ Maintenance or after sales service
- ☒ Entire value chain integration
- ☐ Other

C5.11 Success stories or global best practices for Industrial Automation (provide examples)

Country/Region, key success factors, lessons for Tanzania, and adaptation needs.

SECTION D: INDUSTRY 4.0 AND ADVANCED MANUFACTURING TECHNOLOGIES

D1. Rate the importance of the following automation and robotics technologies for Tanzania's manufacturing and construction sector. Please rate each technology using a scale from 1 to 5, where 1 equals Not important, 2 equals Slightly important, 3 equals Moderately important, 4 equals Very important, and 5 equals Critical.

Automation and Robotics Technologies

	1	2	3	4	5
Industrial robots (welding, assembly, material handling)	☐	☐	☐	☐	☒
Collaborative robots for SMEs	☐	☐	☐	☒	☐
Automated production lines	☐	☐	☐	☐	☒
Computer Numerical Control (CNC) machines	☐	☐	☐	☐	☒

Automated quality inspection systems	☐	☐	☒	☐	☐
Automated warehouse systems (AS or RS)	☐	☐	☐	☒	☐
Programmable Logic Controllers (PLCs)	☐	☐	☐	☒	☐
Automated packaging systems	☐	☐	☐	☒	☐
Pick and place machines	☐	☐	☐	☒	☐
Conveyor automation systems	☐	☐	☐	☒	☐
Other (specify)	☐	☐	☐	☐	☐

D2. Rate the importance of the following digital manufacturing technologies for Tanzania's manufacturing and construction sector using the same 1 to 5 scale (1 equals Not important, 5 equals Critical).

Digital Manufacturing Technologies	1	2	3	4	5
Computer Aided Design (CAD) systems	☐	☐	☐	☐	☐
Computer Aided Manufacturing (CAM) systems	☐	☐	☐	☐	☐
Product Lifecycle Management (PLM) software	☐	☐	☐	☐	☐
Manufacturing Execution Systems (MES)	☐	☐	☐	☐	☐
Enterprise Resource Planning (ERP) for manufacturing	☐	☐	☐	☐	☐
Digital twin technology	☐	☐	☐	☐	☐
Simulation and modelling software	☐	☐	☐	☐	☐
3D printing or additive manufacturing	☐	☐	☐	☐	☐
Virtual prototyping	☐	☐	☐	☐	☐
Building Information Modelling (BIM) for construction	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

D3. Rate the importance of the following smart factory technologies for Tanzania's manufacturing and construction sector using the same 1 to 5 scale (1 equals Not important, 5 equals Critical).

Smart Factory Technologies	1	2	3	4	5
Industrial Internet of Things (IoT) sensors	☐	☐	☐	☒	☐
Real time production monitoring systems	☐	☐	☐	☒	☐
Predictive maintenance systems	☐	☐	☐	☐	☐
Machine to machine (M2M) communication	☐	☐	☐	☐	☐
Cloud based manufacturing platforms	☐	☐	☐	☐	☐

Big data analytics for manufacturing	☐	☐	☐	☐	☐
Artificial Intelligence for quality control	☐	☐	☐	☒	☐
Augmented Reality (AR) for maintenance	☐	☐	☐	☐	☐
Cyber physical systems	☐	☐	☐	☐	☐
Edge computing for factories	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

D4. Industry 4.0 readiness: current level of digital and automation adoption in your sector

- ☐ Manual or traditional methods (0 to 20 percent digitized)
- ☒ Basic mechanization (20 to 40 percent digitized)
- ☐ Partial automation (40 to 60 percent digitized)
- ☐ Advanced automation (60 to 80 percent digitized)
- ☐ Smart factory (80 to 100 percent digitized)

D5. Barriers to Industry 4.0 adoption in Tanzania (select all that apply)

- ☒ High technology costs
- ☒ Lack of awareness or knowledge
- ☒ Shortage of skilled personnel
- ☒ Inadequate ICT infrastructure
- ☒ Limited local technology providers
- ☒ Resistance to change
- ☒ Uncertain return on investment
- ☐ Other

SECTION E: MANUFACTURING SUB-SECTOR SPECIFIC TECHNOLOGIES

E1. Rate the importance of the following technologies for Tanzania's manufacturing sector using a scale from 1 to 5, where 1 equals Not important, 2 equals Slightly important, 3 equals Moderately important, 4 equals Very important, and 5 equals Critical.

Heavy Manufacturing and Metal Industries	1	2	3	4	5
Modern steel production (mini mills, electric arc furnaces)	☐	☐	☐	☐	☐
Aluminum smelting and rolling	☐	☐	☐	☐	☐
Metal casting and foundry technologies	☐	☐	☐	☐	☐
Precision metal cutting and machining	☐	☐	☐	☐	☐
Metal surface treatment (galvanizing, powder coating)	☐	☐	☐	☐	☐

Heavy machinery manufacturing	☐	☐	☐	☐	☐
Industrial equipment fabrication	☐	☐	☐	☐	☐
Shipbuilding technology	☐	☐	☐	☐	☐
Railway equipment manufacturing	☐	☐	☐	☐	☐
Agricultural machinery manufacturing	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E2. Rate the importance of the following technologies for Chemical and Pharmaceutical Manufacturing. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Chemical and Pharmaceutical Manufacturing	1	2	3	4	5
Chemical synthesis and processing equipment	☐	☐	☐	☐	☐
Fertilizer production (NPK blending, urea)	☐	☐	☐	☐	☐
Pesticide formulation facilities	☐	☐	☐	☐	☐
Paint and coating manufacturing	☐	☐	☐	☐	☐
Plastic resin production	☐	☐	☐	☐	☐
Active Pharmaceutical Ingredients (API) production	☐	☐	☐	☐	☐
Generic drug manufacturing	☐	☐	☐	☐	☐
Vaccine production facilities	☐	☐	☐	☐	☐
Quality control laboratories (GMP compliant)	☐	☐	☐	☐	☐
Clean room technology	☐	☐	☐	☐	☐
Bio pharmaceutical manufacturing	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E3. Rate the importance of the following technologies for Food and Beverage Processing (Industrial Scale). Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Food and Beverage Processing (Industrial Scale)	1	2	3	4	5
Modern grain milling (wheat, maize)	☐	☐	☐	☐	☐
Edible oil refining	☐	☐	☐	☐	☐
Sugar refining	☐	☐	☐	☐	☐
Dairy processing (UHT, powdered milk)	☐	☐	☐	☐	☐
Meat processing and cold chain	☐	☐	☐	☐	☐

Fruit juice concentration and bottling	☐	☐	☐	☐	☐
Instant or processed foods manufacturing	☐	☐	☐	☐	☐
Beverage production (soft drinks, water)	☐	☐	☐	☐	☐
Brewery technology	☐	☐	☐	☐	☐
Food packaging technologies	☐	☐	☐	☐	☐
Food safety testing equipment	☐	☐	☐	☐	☐
Halal certification systems	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E4. Rate the importance of the following technologies for Textile and Garment Manufacturing. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Textile and Garment Manufacturing	1	2	3	4	5
Cotton ginning and spinning	☐	☐	☐	☐	☐
Textile weaving and knitting	☐	☐	☐	☐	☐
Fabric dyeing and finishing	☐	☐	☐	☐	☐
Garment cutting and sewing (automated)	☐	☐	☐	☐	☐
Industrial laundry systems	☐	☐	☐	☐	☐
Quality inspection systems	☐	☐	☐	☐	☐
Fashion design software	☐	☐	☐	☐	☐
Textile recycling technology	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E5. Rate the importance of the following technologies for Leather and Footwear. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Leather and Footwear	1	2	3	4	5
Leather tanning and processing	☐	☐	☐	☐	☐
Leather finishing technologies	☐	☐	☐	☐	☐
Footwear manufacturing equipment	☐	☐	☐	☐	☐
Leather goods manufacturing	☐	☐	☐	☐	☐
Quality testing for leather products	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E6. Rate the importance of the following technologies for Electronics and Electrical Equipment. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Electronics and Electrical Equipment	1	2	3	4	5
Printed Circuit Board (PCB) assembly	☐	☐	☐	☐	☐
Electronics assembly lines	☐	☐	☐	☐	☐
Cable and wire manufacturing	☐	☐	☐	☐	☐
Transformer manufacturing	☐	☐	☐	☐	☐
Solar panel assembly	☐	☐	☐	☐	☐
Battery assembly (lithium ion, lead acid)	☐	☐	☐	☐	☐
LED lighting manufacturing	☐	☐	☐	☐	☐
Consumer electronics assembly	☐	☐	☐	☐	☐
Testing and calibration equipment	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E7. Rate the importance of the following technologies for Automotive and Transport Equipment. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Automotive and Transport Equipment	1	2	3	4	5
Vehicle assembly lines	☐	☐	☐	☐	☐
Auto parts manufacturing	☐	☐	☐	☐	☐
Motorcycle and bicycle assembly	☐	☐	☐	☐	☐
Agricultural vehicle assembly	☐	☐	☐	☐	☐
Vehicle body fabrication	☐	☐	☐	☐	☐
Automotive painting facilities	☐	☐	☐	☐	☐
Quality testing (crash test, emissions)	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

E8. Rate the importance of the following technologies for Building Materials Manufacturing. Please use a scale from 1 to 5, where 1 equals Not important and 5 equals Critical.

Building Materials Manufacturing	1	2	3	4	5
Cement production (modern kilns)	☐	☐	☐	☐	☐
Ready mix concrete plants	☐	☐	☐	☐	☐
Concrete block or paver manufacturing	☐	☐	☐	☐	☐

Roofing materials (tiles, sheets)	☐	☐	☐	☐	☐
Steel reinforcement bars (rebar)	☐	☐	☐	☐	☐
Ceramic tile manufacturing	☐	☐	☐	☐	☐
Glass manufacturing	☐	☐	☐	☐	☐
Paint manufacturing	☐	☐	☐	☐	☐
Insulation materials	☐	☐	☐	☐	☐
Prefabricated building components	☐	☐	☐	☐	☐
PVC pipes and fittings	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

SECTION F: CONSTRUCTION TECHNOLOGIES

F1. Modern construction methods priority: Select and rank the top five (5) technologies, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- ☐ Prefabricated or modular construction systems
 ☐ Precast concrete technology
- ☐ Steel frame construction
 ☐ Advanced formwork systems
 ☒ Modern scaffolding systems
- ☐ High rise construction technologies
 ☐ Bridge construction methods (cable stayed, suspended)
- ☐ Tunnel boring machines
 ☐ Road construction equipment (asphalt plants, pavers)
- ☐ Concrete pumping technology
 ☐ Other

Rank 2

- ☐ Prefabricated or modular construction systems
 ☐ Precast concrete technology
- ☒ Steel frame construction
 ☐ Advanced formwork systems
 ☐ Modern scaffolding systems
- ☐ High rise construction technologies
 ☐ Bridge construction methods (cable stayed, suspended)
- ☐ Tunnel boring machines
 ☐ Road construction equipment (asphalt plants, pavers)
- ☐ Concrete pumping technology
 ☐ Other

Rank 3

- ☐ Prefabricated or modular construction systems
 ☐ Precast concrete technology
- ☐ Steel frame construction
 ☒ Advanced formwork systems
 ☐ Modern scaffolding systems
- ☐ High rise construction technologies
 ☐ Bridge construction methods (cable stayed, suspended)
- ☐ Tunnel boring machines
 ☐ Road construction equipment (asphalt plants, pavers)
- ☐ Concrete pumping technology
 ☐ Other

Rank 4

- ☐ Prefabricated or modular construction systems
 ☒ Precast concrete technology
 ☐ Steel frame construction
 ☐ Advanced formwork systems
 ☐ Modern scaffolding systems
 ☐ High rise construction technologies
 ☐ Bridge construction methods (cable stayed, suspended)
 ☐ Tunnel boring machines
 ☐ Road construction equipment (asphalt plants, pavers)
 ☐ Concrete pumping technology
 ☐ Other

Rank 5

- ☐ Prefabricated or modular construction systems
 ☐ Precast concrete technology
 ☐ Steel frame construction
 ☐ Advanced formwork systems
 ☐ Modern scaffolding systems
 ☒ High rise construction technologies
 ☐ Bridge construction methods (cable stayed, suspended)
 ☐ Tunnel boring machines
 ☐ Road construction equipment (asphalt plants, pavers)
 ☐ Concrete pumping technology
 ☐ Other

F2. Green and sustainable construction: Select and rank the top five (5) technologies, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- ☐ Green building certification systems (LEED, BREEAM adaptation)
 ☒ Energy efficient building designs
 ☐ Solar integration in buildings (BIPV)
 ☐ Rainwater harvesting systems
 ☐ Greywater recycling systems
 ☐ Natural ventilation technologies
 ☐ Green roofs and walls
 ☐ Sustainable building materials (bamboo, earth blocks)
 ☐ Low carbon concrete
 ☐ Recycled building materials
 ☐ Energy modelling software
 ☐ Other

Rank 2

- ☐ Green building certification systems (LEED, BREEAM adaptation)
 ☐ Energy efficient building designs
 ☐ Solar integration in buildings (BIPV)
 ☒ Rainwater harvesting systems
 ☐ Greywater recycling systems
 ☐ Natural ventilation technologies
 ☐ Green roofs and walls
 ☐ Sustainable building materials (bamboo, earth blocks)
 ☐ Low carbon concrete
 ☐ Recycled building materials
 ☐ Energy modelling software
 ☐ Other

Rank 3

- ☐ Green building certification systems (LEED, BREEAM adaptation)
 ☐ Energy efficient building designs
 ☐ Solar integration in buildings (BIPV)
 ☐ Rainwater harvesting systems
 ☒ Greywater recycling systems
 ☐ Natural ventilation technologies
 ☐ Green roofs and walls
 ☐ Sustainable building materials (bamboo, earth blocks)
 ☐ Low carbon concrete
 ☐ Recycled building materials
 ☐ Energy modelling software
 ☐ Other

Rank 4

- ☒ Green building certification systems (LEED, BREEAM adaptation)
 ☐ Energy efficient building designs
☐ Solar integration in buildings (BIPV)
 ☐ Rainwater harvesting systems
☐ Greywater recycling systems
 ☐ Natural ventilation technologies
 ☐ Green roofs and walls
☐ Sustainable building materials (bamboo, earth blocks)
 ☐ Low carbon concrete
☐ Recycled building materials
 ☐ Energy modelling software
 ☐ Other

Rank 5

- ☐ Green building certification systems (LEED, BREEAM adaptation)
 ☐ Energy efficient building designs
☐ Solar integration in buildings (BIPV)
 ☐ Rainwater harvesting systems
☐ Greywater recycling systems
 ☒ Natural ventilation technologies
 ☐ Green roofs and walls
☐ Sustainable building materials (bamboo, earth blocks)
 ☐ Low carbon concrete
☐ Recycled building materials
 ☐ Energy modelling software
 ☐ Other

F3. Smart building technologies: Select and rank the top five (5) technologies, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- ☐ Building Management Systems (BMS)
 ☒ Smart lighting systems
☐ HVAC automation
 ☐ Access control systems
☐ Fire detection and suppression systems
 ☐ Energy monitoring systems
☐ Smart elevators
 ☐ Integrated security systems
 ☐ IoT sensors for buildings
☐ Smart parking systems
 ☐ Other

Rank 2

- ☐ Building Management Systems (BMS)
 ☐ Smart lighting systems
☒ HVAC automation
 ☐ Access control systems
☐ Fire detection and suppression systems
 ☐ Energy monitoring systems
☐ Smart elevators
 ☐ Integrated security systems
 ☐ IoT sensors for buildings
☐ Smart parking systems
 ☐ Other

Rank 3

- ☐ Building Management Systems (BMS)
 ☐ Smart lighting systems
☐ HVAC automation
 ☒ Access control systems
☐ Fire detection and suppression systems
 ☐ Energy monitoring systems
☐ Smart elevators
 ☐ Integrated security systems
 ☐ IoT sensors for buildings
☐ Smart parking systems
 ☐ Other

Rank 4

- ☐ Building Management Systems (BMS)
 ☐ Smart lighting systems
☐ HVAC automation
 ☐ Access control systems
☐ Fire detection and suppression systems
 ☒ Energy monitoring systems
☐ Smart elevators
 ☐ Integrated security systems
 ☐ IoT sensors for buildings
☐ Smart parking systems
 ☐ Other

Rank 5

- ☒ Building Management Systems (BMS)
 ☐ Smart lighting systems
☐ HVAC automation
 ☐ Access control systems
☐ Fire detection and suppression systems
 ☐ Energy monitoring systems
☐ Smart elevators
 ☐ Integrated security systems
 ☐ IoT sensors for buildings
☐ Smart parking systems
 ☐ Other

F4. Construction equipment and machinery: critical equipment needs (select all that apply)

- ☐ Excavators and earth moving equipment
☐ Concrete mixers and batching plants
☐ Tower cranes
☐ Mobile cranes
☐ Bulldozers and graders
☐ Compactors and rollers
☐ Piling equipment
☐ Concrete pumps
☐ Generators
☐ Welding equipment
☐ Material handling equipment
☐ Survey equipment (total stations, GPS, drones)
☐ Other

F5. Building Information Modeling (BIM): current BIM adoption in Tanzania

- ☐ Not used
☐ Pilot projects only
☐ Limited adoption (less than 20 percent of projects)
☐ Moderate adoption (20 to 50 percent)
☐ Widespread adoption (more than 50 percent)

F5.1 BIM implementation barriers (select all that apply)

- ☐ Lack of skilled personnel
- ☐ Software costs
- ☐ Limited awareness
- ☐ Client demand insufficient
- ☐ Inadequate training institutions
- ☐ Internet connectivity issues
- ☐ Other

F6. Affordable housing construction: technologies for mass housing delivery (select top 5)

- ☐ Interlocking stabilized soil blocks (ISSB)
- ☐ Prefabricated panel systems
- ☐ 3D concrete printing
- ☐ Modular housing
- ☐ Light steel frame systems
- ☐ Compressed earth blocks
- ☐ Low cost roofing systems
- ☐ Self help housing technologies
- ☐ Community based construction methods
- ☐ Incremental housing systems
- ☐ Other

SECTION G: SMART CITIES AND URBAN INFRASTRUCTURE

G1. Rate the importance of the following smart city technologies for Tanzania (1-5): 1=Not important, 2=Slightly important, 3=Important, 4=Very important, 5=Critical.

Smart city technologies	1	2	3	4	5
Smart traffic management systems	☐	☐	☐	☐	☐
Intelligent street lighting	☐	☐	☐	☐	☐
Smart waste management (sensors, route optimization)	☐	☐	☐	☐	☐
Smart water supply management	☐	☐	☐	☐	☐
Smart sewerage and drainage systems	☐	☐	☐	☐	☐
Urban air quality monitoring	☐	☐	☐	☐	☐
Smart public transport systems	☐	☐	☐	☐	☐
Digital city planning tools (GIS, 3D modeling)	☐	☐	☐	☐	☐

Smart parking solutions	☐	☐	☐	☐	☐
Public Wi Fi infrastructure	☐	☐	☐	☐	☐
E governance platforms	☐	☐	☐	☐	☐
Smart energy grids for cities	☐	☐	☐	☐	☐
Integrated command and control centers	☐	☐	☐	☐	☐
Other	☐	☐	☐	☐	☐

G2. Rate the importance of the following urban planning technologies for Tanzania (1-5): 1=Not important, 2=Slightly important, 3=Important, 4=Very important, 5=Critical.

Urban planning technologies	1	2	3	4	5
Geographic Information Systems (GIS)	☐	☐	☐	☐	☐
Remote sensing and satellite imagery	☐	☐	☐	☐	☐
3D city modeling	☐	☐	☐	☐	☐
Urban growth simulation software	☐	☐	☐	☐	☐
Land use planning tools	☐	☐	☐	☐	☐
Infrastructure network modeling	☐	☐	☐	☐	☐
Population density analysis tools	☐	☐	☐	☐	☐
Climate responsive urban design tools	☐	☐	☐	☐	☐
Other	☐	☐	☐	☐	☐

SECTION H: QUALITY CONTROL AND STANDARDS

H1. Testing and certification infrastructure (select all that apply)

- ☐ Materials testing laboratories (metals, concrete, etc.)
- ☐ Chemical analysis laboratories
- ☐ Microbiological testing labs
- ☐ Food safety testing labs
- ☐ Pharmaceutical quality control labs
- ☐ Textile testing facilities
- ☐ Electronics testing and calibration
- ☐ Environmental testing (emissions, effluents)
- ☐ Product safety testing
- ☐ Metrology and standards labs
- ☐ Non destructive testing facilities
- ☐ Mobile testing laboratories
- ☐ Other (specify)

H2. Quality management systems (select all that apply)

- ☐ ISO 9001 (Quality Management)
- ☐ ISO 14001 (Environmental Management)
- ☐ ISO 45001 (Occupational Health and Safety)
- ☐ ISO 50001 (Energy Management)
- ☐ Good Manufacturing Practice (GMP)
- ☐ HACCP (Food Safety)
- ☐ Six Sigma methodologies
- ☐ Total Quality Management (TQM) systems
- ☐ Statistical Process Control (SPC)
- ☐ Other (specify)

SECTION I: QUALITY, STANDARDS AND TESTING

I1. Critical infrastructure in industrial parks: Select and rank the top five (5) services, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- | | | |
|--|--|--|
| ☐ Reliable electricity supply (24/7) | ☐ Water supply and treatment | ☒ Wastewater treatment plants |
| ☐ Natural gas pipeline | ☐ Fiber optic connectivity | ☐ Access roads and internal roads |
| ☐ Rail siding | ☐ Common effluent treatment plants | |
| ☐ Waste management facilities | ☐ Firefighting systems | ☐ Security systems |
| ☐ Common testing facilities | ☐ Worker housing/dormitories | ☐ Canteen and welfare facilities |
| ☐ Other | | |

Rank 2

- | | | |
|---|--|---|
| ☒ Reliable electricity supply (24/7) | ☐ Water supply and treatment | ☐ Wastewater treatment plants |
| ☐ Natural gas pipeline | ☐ Fiber optic connectivity | ☐ Access roads and internal roads |
| ☐ Rail siding | ☐ Common effluent treatment plants | |
| ☐ Waste management facilities | ☐ Firefighting systems | ☐ Security systems |
| ☐ Common testing facilities | ☐ Worker housing/dormitories | ☐ Canteen and welfare facilities |
| ☐ Other | | |

Rank 3

- | | | |
|--|--|---|
| ☐ Reliable electricity supply (24/7) | ☐ Water supply and treatment | ☐ Wastewater treatment plants |
| ☒ Natural gas pipeline | ☐ Fiber optic connectivity | ☐ Access roads and internal roads |
| ☐ Rail siding | ☐ Common effluent treatment plants | |
| ☐ Waste management facilities | ☐ Firefighting systems | ☐ Security systems |
| ☐ Common testing facilities | ☐ Worker housing/dormitories | ☐ Canteen and welfare facilities |
| ☐ Other | | |

Rank 4

- | | | |
|--|---|---|
| ☐ Reliable electricity supply (24/7) | ☐ Water supply and treatment | ☐ Wastewater treatment plants |
| ☐ Natural gas pipeline | ☒ Fiber optic connectivity | ☐ Access roads and internal roads |
| ☐ Rail siding | ☐ Common effluent treatment plants | |
| ☐ Waste management facilities | ☐ Firefighting systems | ☐ Security systems |
| ☐ Common testing facilities | ☐ Worker housing/dormitories | ☐ Canteen and welfare facilities |
| ☐ Other | | |

Rank 5

- | | | |
|--|--|--|
| ☐ Reliable electricity supply (24/7) | ☐ Water supply and treatment | ☐ Wastewater treatment plants |
| ☐ Natural gas pipeline | ☐ Fiber optic connectivity | ☒ Access roads and internal roads |
| ☐ Rail siding | ☐ Common effluent treatment plants | |
| ☐ Waste management facilities | ☐ Firefighting systems | ☐ Security systems |
| ☐ Common testing facilities | ☐ Worker housing/dormitories | ☐ Canteen and welfare facilities |
| ☐ Other | | |

12. Industrial Support Services: Select and rank the top five (5) services, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- | | | |
|---|---|---|
| ☐ Tool and die making workshops | ☐ Industrial maintenance services | ☒ Calibration services |
| ☐ Industrial machinery repair | ☐ Spare parts supply networks | |
| ☐ Industrial gas supply (oxygen, nitrogen, acetylene) | ☐ Industrial chemicals supply | |
| ☐ Logistics and warehousing services | ☐ Packaging materials supply | |
| ☐ Industrial waste management services | ☐ Other | |

Rank 2

- | | | |
|---|---|--|
| ☒ Tool and die making workshops | ☐ Industrial maintenance services | ☐ Calibration services |
| ☐ Industrial machinery repair | ☐ Spare parts supply networks | |
| ☐ Industrial gas supply (oxygen, nitrogen, acetylene) | ☐ Industrial chemicals supply | |
| ☐ Logistics and warehousing services | ☐ Packaging materials supply | |
| ☐ Industrial waste management services | ☐ Other | |

Rank 3

- | | | |
|---|--|--|
| ☐ Tool and die making workshops | ☒ Industrial maintenance services | ☐ Calibration services |
| ☐ Industrial machinery repair | ☐ Spare parts supply networks | |
| ☐ Industrial gas supply (oxygen, nitrogen, acetylene) | ☐ Industrial chemicals supply | |
| ☐ Logistics and warehousing services | ☐ Packaging materials supply | |
| ☐ Industrial waste management services | ☐ Other | |

Rank 4

- | | | |
|---|--|--|
| ☐ Tool and die making workshops | ☐ Industrial maintenance services | ☐ Calibration services |
| ☐ Industrial machinery repair | ☐ Spare parts supply networks | |
| ☐ Industrial gas supply (oxygen, nitrogen, acetylene) | ☒ Industrial chemicals supply | |
| ☐ Logistics and warehousing services | ☐ Packaging materials supply | |
| ☐ Industrial waste management services | ☐ Other | |

Rank 5

- ☐ Tool and die making workshops ☐ Industrial maintenance services ☐ Calibration services
- ☐ Industrial machinery repair ☐ Spare parts supply networks
- ☒ Industrial gas supply (oxygen, nitrogen, acetylene) ☐ Industrial chemicals supply
- ☐ Logistics and warehousing services ☐ Packaging materials supply
- ☐ Industrial waste management services ☐ Other

SECTION J: ENERGY FOR MANUFACTURING AND CONSTRUCTION**J1. Industrial Energy Requirements: Current energy challenges: (Select all that apply)**

- ☐ Frequent power outages
- ☐ High electricity tariffs
- ☒ Insufficient grid capacity
- ☐ Limited three-phase supply
- ☒ No natural gas access
- ☒ High cost of backup generators
- ☐ Load shedding/rationing
- ☒ Poor power quality (voltage fluctuations)

J2. Alternative Energy for Industry: (Select all that apply)

- ☒ Industrial solar PV systems
- ☐ Biomass-based power generation
- ☒ Natural gas for manufacturing
- ☐ Industrial energy storage (batteries)
- ☐ Combined Heat and Power (CHP) systems
- ☐ Waste-to-energy systems
- ☐ Industrial energy efficiency audits
- ☐ Energy management systems (ISO 50001)
- ☐ Power factor correction equipment

J3. Energy Intensity: What is acceptable energy cost as % of production cost?

- ☐ <5%
- ☐ 5-10%
- ☐ 10-20%
- ☐ 20-30%
- ☐ >30%

SECTION K: RESEARCH & DEVELOPMENT FOR MANUFACTURING

K1. Industrial R&D Infrastructure: Critical R&D facilities needed: (Select top 5)

- ☐ Industrial design centers
- ☐ Prototyping facilities
- ☐ Pilot manufacturing plants
- ☐ Technology demonstration centers
- ☐ Testing and certification laboratories
- ☐ Materials research laboratories
- ☐ Industrial automation research center
- ☐ Packaging development center
- ☐ Product development workshops
- ☐ Failure analysis laboratories
- ☐ Reverse engineering facilities

K2. University-Industry Linkages: How can universities better support manufacturing?

Imparting relevant curriculum with more practises with modern teaching equipment; And advocating for manufacturers to accept internships to enable students equipped with the current technology .

K3. Technology Adaptation and Localization: Rate each item from 1-5 (1 = Not important, 5 = Critical).

Technology Adaptation and Localization	1	2	3	4	5
Technology needs assessment capabilities	☐	☐	☐	☐	☐
Technology adaptation/modification facilities	☐	☐	☐	☐	☐
Indigenous technology development	☐	☐	☐	☐	☐
Appropriate technology research	☐	☐	☐	☐	☐
Technology documentation and dissemination	☐	☐	☐	☐	☐

SECTION L: SKILLS DEVELOPMENT FOR MANUFACTURING & CONSTRUCTION

L1. Critical Skills Gaps: What are the THREE most critical skills shortages in Manufacturing & Construction?

1.

Engineering and technical craft

2.

Technical

3.

Automation

L2. Priority Training Areas - Manufacturing: Rate the importance of each area on a scale of 1–5 (1 = Not important, 5 = Critical).

Priority Training Areas - Manufacturing	1	2	3	4	5
CNC machine operation and programming	☐	☐	☐	☐	☐
Industrial automation and PLC programming	☐	☐	☐	☐	☐
Industrial robotics	☐	☐	☐	☐	☐
Quality control and testing	☐	☐	☐	☐	☐
Industrial maintenance	☐	☐	☐	☐	☐
Welding and metal fabrication	☐	☐	☐	☐	☐
Industrial electrical systems	☐	☐	☐	☐	☐
Instrumentation and control	☐	☐	☐	☐	☐
Lean manufacturing and Six Sigma	☐	☐	☐	☐	☐
Industrial safety and OSHA standards	☐	☐	☐	☐	☐
Production planning and inventory control	☐	☐	☐	☐	☐
Industrial design and CAD/CAM	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐
Priority Training Areas - Construction	1	2	3	4	5
Modern construction methods	☐	☐	☐	☐	☐
BIM and digital construction	☐	☐	☐	☐	☐
Project management (PRINCE2, PMP)	☐	☐	☐	☐	☐
Quantity surveying	☐	☐	☐	☐	☐
Construction equipment operation	☐	☐	☐	☐	☐
Green building techniques	☐	☐	☐	☐	☐
Structural engineering	☐	☐	☐	☐	☐
MEP (Mechanical, Electrical, Plumbing)	☐	☐	☐	☐	☐

Construction safety management	☐	☐	☐	☐	☐
Cost estimation and control	☐	☐	☐	☐	☐
Contract management	☐	☐	☐	☐	☐
Other (specify)	☐	☐	☐	☐	☐

L3. Training Infrastructure Needs: Rate the importance of each item on a scale of 1–5 (1 = Not important, 5 = Critical).

Training Infrastructure Needs	1	2	3	4	5
Upgrade VETA training centers with modern equipment	☐	☐	☐	☐	☒
Establish Industrial Training Institutes (ITIs)	☐	☐	☐	☐	☒
Set up company-based training centers	☐	☐	☒	☐	☐
Develop apprenticeship programs	☐	☐	☐	☒	☐
Establish Centers of Excellence for specific technologies	☐	☐	☐	☒	☐
Online/e-learning platforms for technical training	☐	☐	☐	☒	☐
Train-the-trainer programs	☐	☐	☐	☒	☐
International exchange programs	☐	☐	☐	☒	☐
Industry certification programs	☐	☐	☐	☒	☐
Other (specify)	☐	☐	☐	☐	☐

SECTION M: FINANCING MANUFACTURING AND CONSTRUCTION TECHNOLOGY

M1. Current Financing Challenges (Select all that apply)

- ☒ High interest rates on industrial loans
- ☐ Short loan tenors (inadequate for long-term projects)
- ☒ Collateral requirements too strict
- ☐ Limited understanding of manufacturing by banks
- ☐ No specialized industrial finance institutions
- ☒ High currency exchange risk for imported equipment
- ☐ Limited access to venture capital
- ☐ Inadequate government support programs
- ☐ Complex application procedures
- ☐ Lack of risk capital for new technologies
- ☐ Other (specify)

M2. Preferred Financing Mechanisms (Select all that apply)

- ☐ Development Finance Institutions (DFIs)
- ☐ Industrial Development Bank
- ☐ Commercial bank industrial loans
- ☐ Equipment leasing/hire purchase
- ☐ Vendor financing
- ☐ Private equity investment
- ☐ Venture capital for manufacturing startups
- ☐ Government matching grants
- ☐ Special Economic Zone incentives
- ☐ Export credit agencies
- ☐ PPPs for industrial infrastructure
- ☐ Green bonds for sustainable manufacturing
- ☐ Diaspora investment
- ☐ Other (specify)

M3. Investment Incentives Needed (Select all that apply)

- ☐ Tax holidays for new industries
- ☒ Duty exemptions on capital equipment
- ☐ Subsidized industrial land
- ☐ Reduced electricity tariffs for industry
- ☒ R&D tax credits
- ☒ Export incentives
- ☐ Training grants
- ☐ Technology upgrade grants
- ☒ Green technology incentives
- ☐ Other (specify)

SECTION N: POLICY & REGULATORY FRAMEWORK

N1. POLICY REFORMS NEEDED (Select top 5)

- ☒ Industrial policy clarity and consistency
- ☒ Local content requirements
- ☐ Import tariff rationalization
- ☐ Export promotion policies
- ☒ Competition policy (against dumping)
- ☐ Technology transfer regulations
- ☐ Industrial land allocation policy
- ☐ Energy tariff policy for industry
- ☐ Labor laws (flexibility vs protection)
- ☒ Environmental regulations (compliance costs)
- ☒ Foreign investment regulations
- ☐ Standards and quality regulations
- ☐ Construction industry regulations
- ☐ Urban planning regulations
- ☐ Special Economic Zone regulations
- ☐ Other (specify)

N2. INSTITUTIONAL STRENGTHENING: Which institutions need strengthening? (Select all that apply)

- ☒ Tanzania Industrial Research and Development Organization (TIRDO)
- ☒ Small Industries Development Organization (SIDO)
- ☒ Tanzania Bureau of Standards (TBS)
- ☒ Weights and Measures Agency
- ☒ VETA and technical training institutes
- ☒ Export Processing Zones Authority (EPZA)
- ☒ National Construction Council
- ☒ Engineers Registration Board
- ☒ Architects Registration Board
- ☒ University engineering departments
- ☒ Industrial associations and chambers
- ☒ Contractor Development Board
- ☒ Other (specify)

N2. Other institution needing strengthening (specify)

Fair Competition Commission

N3. REGULATORY BARRIERS: Key regulatory barriers to technology adoption:

SECTION O: EXPORT MANUFACTURING & COMPETITIVENESS**O1. EXPORT MARKETS: Target export markets for manufactured goods (Select all that apply)**

- ☐ East African Community (EAC)
- ☐ Southern African Development Community (SADC)
- ☐ African Continental Free Trade Area (AfCFTA)
- ☐ Middle East
- ☐ Europe
- ☐ North America
- ☐ Asia
- ☐ Other (specify)

O2. EXPORT COMPETITIVENESS FACTORS: Select and rank the top five (5) factors, where Rank 1 = most urgent or highest priority, Rank 2 = very high priority, Rank 3 = high priority, Rank 4 = moderate priority, and Rank 5 = lower priority among the selected five.

Rank 1

- ☐ Production costs
- ☐ Product quality issues
- ☐ Product variety
- ☐ Consistent supply
- ☐ Logistics costs
- ☐ Market knowledge
- ☐ Trade financing
- ☒ Non-tariff barriers in export markets
- ☐ Compliance with international standards
- ☐ Brand/reputation issues
- ☐ Other (specify)

Rank 2

- ☐ Production costs
- ☐ Product quality issues
- ☐ Product variety
- ☐ Consistent supply
- ☐ Logistics costs
- ☐ Market knowledge
- ☐ Trade financing
- ☐ Non-tariff barriers in export markets
- ☐ Compliance with international standards
- ☐ Brand/reputation issues
- ☐ Other (specify)

Rank 3

- ☐ Production costs
- ☐ Product quality issues
- ☐ Product variety
- ☐ Consistent supply
- ☐ Logistics costs
- ☐ Market knowledge
- ☐ Trade financing
- ☐ Non-tariff barriers in export markets
- ☐ Compliance with international standards
- ☐ Brand/reputation issues
- ☐ Other (specify)

Rank 4

- ☐ Production costs
- ☐ Product quality issues
- ☐ Product variety
- ☐ Consistent supply
- ☐ Logistics costs
- ☐ Market knowledge
- ☐ Trade financing
- ☐ Non-tariff barriers in export markets
- ☐ Compliance with international standards
- ☐ Brand/reputation issues
- ☐ Other (specify)

Rank 5

- ☐ Production costs
- ☐ Product quality issues
- ☐ Product variety
- ☐ Consistent supply
- ☐ Logistics costs
- ☐ Market knowledge
- ☐ Trade financing
- ☐ Non-tariff barriers in export markets
- ☐ Compliance with international standards
- ☐ Brand/reputation issues
- ☐ Other (specify)

03. GLOBAL VALUE CHAIN INTEGRATION: How can manufacturing integrate into global value chains?**SECTION P: SUSTAINABLE MANUFACTURING & CIRCULAR ECONOMY**

P1. GREEN MANUFACTURING TECHNOLOGIES: Rate the importance of each item from 1–5 (1 = Not important, 5 = Critical).

Green Manufacturing Technologies	1	2	3	4	5
Energy-efficient manufacturing equipment	☐	☐	☐	☐	☒
Waste heat recovery systems	☐	☐	☒	☐	☐
Water recycling and reuse systems	☐	☐	☐	☐	☒

Industrial wastewater treatment	☐	☐	☐	☐	☒
Clean production technologies	☐	☐	☐	☒	☐
Renewable energy integration	☐	☐	☐	☒	☐
Carbon footprint measurement and reduction	☐	☐	☐	☒	☐
Life cycle assessment (LCA) tools	☐	☐	☒	☐	☐
Green supply chain management	☐	☐	☐	☒	☐
Other (specify)	☐	☐	☐	☐	☐

P2. CIRCULAR ECONOMY TECHNOLOGIES (Select all that apply)

- ☒ Industrial waste recycling facilities
- ☒ Plastic recycling and reprocessing
- ☒ Metal scrap recycling
- ☒ E-waste recycling
- ☐ Construction and demolition waste recycling
- ☒ Industrial symbiosis platforms
- ☒ By-product utilization technologies
- ☐ Remanufacturing systems
- ☒ Product take-back systems
- ☐ Other (specify)

P3. ENVIRONMENTAL COMPLIANCE: Current environmental management capabilities:

- ☐ No environmental management system
- ☐ Basic compliance with regulations
- ☒ ISO 14001 certified
- ☐ Advanced environmental performance
- ☐ Zero waste/circular economy model
- ☐ Other (specify)

SECTION Q: DIGITALIZATION & SMART MANUFACTURING

Q1. DIGITAL INFRASTRUCTURE FOR MANUFACTURING (Select all that apply)

- ☐ Industrial broadband connectivity
- ☐ Cloud computing for manufacturing
- ☐ Cybersecurity for industrial systems
- ☐ Data centers for industrial data
- ☐ 5G for industrial applications
- ☐ Industrial IoT platforms
- ☐ Manufacturing analytics platforms
- ☐ Supply chain visibility systems
- ☐ E-procurement systems
- ☐ Digital payment systems for B2B
- ☐ Other (specify)

Q2. DIGITAL SKILLS FOR MANUFACTURING (Select all that apply)

- ☐ Data analytics for manufacturing
- ☐ Industrial cybersecurity
- ☐ IIoT system integration
- ☐ MES and ERP systems
- ☐ CAD/CAM/CAE software
- ☐ Simulation and modeling
- ☐ Additive manufacturing design
- ☐ Digital twin development
- ☐ AR/VR for training and maintenance
- ☐ Other (specify)

SECTION S: INTERNATIONAL COOPERATION

S1. STRATEGIC TECHNOLOGY PARTNERS: Best countries/regions for manufacturing technology partnerships (Select all that apply)

- ☐ China (affordable technology, large-scale manufacturing)
- ☐ India (SME-appropriate technology)
- ☐ South Korea (electronics, automotive)
- ☐ Japan (precision engineering, quality systems)
- ☐ Germany (engineering, Industry 4.0)
- ☐ Italy (fashion, design, machinery)
- ☐ Turkey (textile, construction)
- ☐ Thailand (automotive, food processing)
- ☐ Vietnam (garment, light manufacturing)
- ☐ South Africa (mining equipment, construction)
- ☐ Ethiopia (textile, leather)
- ☐ Kenya (manufacturing, construction)
- ☐ Other (specify)

S2. TECHNOLOGY TRANSFER MECHANISMS (Select all that apply)

- ☐ Joint ventures with foreign manufacturers
- ☐ Technology licensing agreements
- ☐ Franchising arrangements
- ☐ Foreign Direct Investment (FDI) with technology transfer clauses
- ☐ Technical assistance projects
- ☐ Equipment supplier partnerships
- ☐ Industrial twinning programs
- ☐ Export processing zones attracting foreign companies
- ☐ Training and capacity building programs
- ☐ Research collaborations
- ☐ Other (specify)

S3. DIASPORA CONTRIBUTION: How can Tanzanian diaspora support manufacturing development?

By providing capital investment and technology transfer.

SECTION T: ADDITIONAL INSIGHTS – MANUFACTURING AND CONSTRUCTION FOCUS

T1. What is the single biggest opportunity for technological transformation in Tanzania's manufacturing and construction sector that is currently under exploited

Automation and data-driven smart system.

T2. What is the single biggest threat to successful manufacturing and construction technology roadmap implementation

Shortage of technical and digital skills and readiness to embrace changes.

Country

TANZANIA

Transformation achieved

CTI

Key technologies used

Success factors

Lessons for Tanzania

Timeline of transformation
